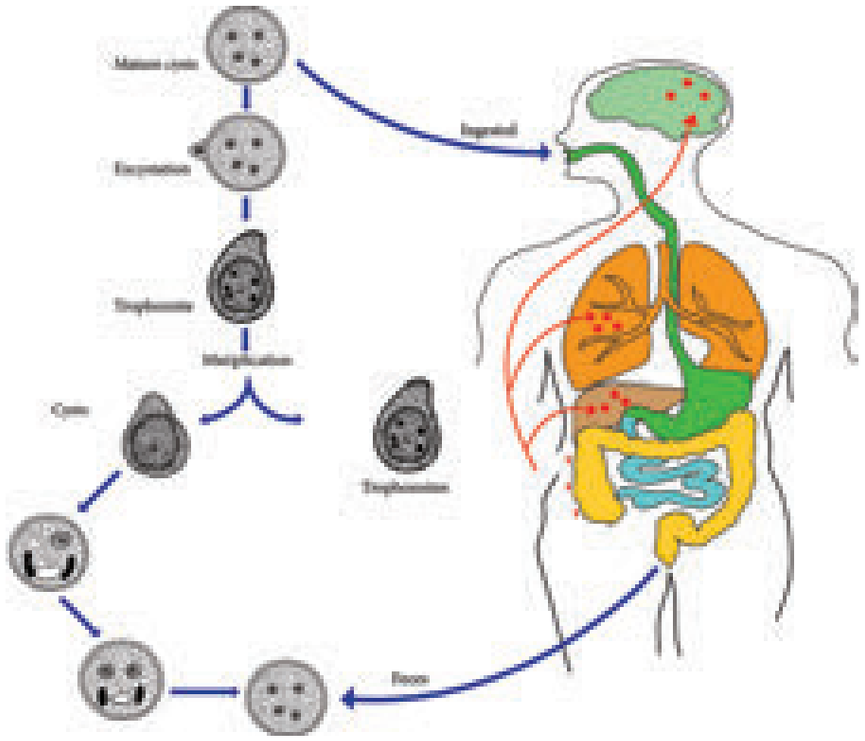


Figure 8.2: Life cycle of *Entamoeba histolytica*

- Some of the trophozoites in colon develop into precystic forms and cysts, which are passed in feces to repeat the cycle.

8.2.5 Pathogenesis

E. histolytica causes intestinal and extra intestinal amoebiasis (Flowchart 8.3).

E. histolytica can live in the intestine without causing symptoms. But, they can also cause severe disease. These amoebas may invade the wall of the intestine leading to amoebic dysentery, an illness that causes intestinal ulcers, bleeding, increased mucus production and diarrhoea. The ulcers are strictly confined to the large intestine being most numerous in the caecum and next in the sigmoid-rectal regions. The lesions may be generalized or localised. A typical amoebic ulcer varies from pin's head to one inch or more in diameter in size. The shape of ulcer may be round or oval. On vertical section, the ulcer appears like flask, with mouth and neck being narrow and base being large and rounded (Figure 8.3 shows the flask – shaped ulcer). The base of ulcer is generally formed by the muscular coat and filled up by the necrotic material. The ulcers generally do not extend deeper than submucosal layer.

8.2.6 Clinical Features

Incubation period is highly variable, but is generally 4 to 5 days.

A wide spectrum, from asymptomatic infection (luminal amoebiasis), to invasive intestinal amoebiasis (dysentery, colitis, appendicitis, toxic mega colon, amoebomas), to invasive extraintestinal amoebiasis occurs. Flowchart 8.4 classifies the clinical outcomes of infection with *Entamoeba histolytica*. Only about 10% to 20% of people who are infected with *E. histolytica* become sick from the infection.

TN_GOVT_XII_Micro_Biology_CH08.indd 159 27-02-2019 14:52:59